Task Decomposition & Specs

Project: AI Tool Comparison Guide Agent

1.What is Task Decomposition?

It simply means — breaking the agent's big job into small steps. Each step has a clear input (what goes in) and output (what comes out).

2.The Agent's Full Job

Take user's comparison criteria → collect tool data → analyse features → score & rank tools → generate comparison report → display results

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Task 1—Collect User Input

What happens: The user fills a form specifying which AI tools to compare and their priorities.

Input (what the user provides):-

• Tool category (e.g. AI Writing Tools, AI Image Generators, AI Coding Assistants)

• Number of tools to compare (e.g. 3–5) • Key criteria to evaluate (e.g. pricing, ease of use, output quality, integrations)

• Use case / audience (e.g. freelancers, enterprise teams, students) Output (passed to next step)

• A structured data object with all 4 fields filled

• Criteria ranked by user priority Decision Points- If any required field is empty → show error, don't proceed- If fewer than 2 tools selected → show warning: 'Need at least 2 tools to compare'

Task 2—Identify & Shortlist AI Tools

What happens: The agent reads the category and use case to identify the most relevant AI tools available for comparison.

Input:-

• Tool category from Task 1

• Use case / audience

• Number of tools requested Output

• A shortlist of tools with names, official URLs, and brief descriptions

• Example: ChatGPT, Claude, Gemini (for AI Writing category)

Decision Points:-

- If fewer tools exist than requested → use all available and notify user

- If category is ambiguous → agent picks the closest match and explains its choice.

Task 3—Research Tool Features (Web Search)

What happens: For each shortlisted tool, the agent searches online to gather up-to-date information on pricing, features, and ratings.

Input:-

• One search query per tool

• Example query: "ChatGPT pricing plans features 2025"

• User-defined criteria list (pricing, integrations, ease of use, etc.) Output

• Feature data per tool mapped to each evaluation criterion

• Pricing tiers (Free / Pro / Enterprise)

• Top 2–3 review sources or ratings per tool

Decision Points:-

- If no results found for a tool → agent uses its own built-in knowledge instead

- If pricing info seems outdated → flag it as 'Verify before purchase

Task 4—Score & Rank Tools

What happens: The agent scores each tool against the user's criteria and produces a ranked comparison.

Input:-

• Feature data from Task 3

• Criteria priority list from Task 1

Criterion	***Max Score***
Output Quality	1.5
Pricing / Value	1.25
Ease of Use	1.0
Integrations	0.75
Customer Support	0.5
	Total 5

Output

• Each tool scored out of 5 across all criteria

• A ranked leaderboard: 1st, 2nd, 3rd place with scores

• Example: Claude — 4.3 | ChatGPT — 4.1 | Gemini — 3.9

Decision Points

- If two tools tie → rank by the highest-priority criterion as tiebreaker

- If no data exists for a criterion → assign 0 and flag as 'No data available

Task 5 — Generate Comparison Report

What happens: The agent compiles all data into a structured, human-readable report with a clear recommendation.

Input

• Ranked scores from Task 4

• Feature data from Task 3

• Original user input — category, criteria, use case

Output

• Side-by-side comparison table (tools vs criteria)

• Pros & Cons section per tool (3 bullet points each)

• A 'Best Pick' recommendation with a short justification

• A 'Best Budget Option' callout if pricing was a top criterion

Decision Points

- If score difference between top 2 tools is under 5 pts → label it 'Too close to call'

- Always end with disclaimer: 'Prices and features may change — verify before purchase

Task 6 — Evaluate Report Quality (LLM-as-Judge)

What happens: A second LLM call reads the generated report and scores it on a fixed rubric before showing it to the user.

Input

• The full comparison report from Task 5

• The original user input (category, tools, criteria, use case)

Criteria	***Max Score***
Are all selected tools covered?	3
Is each user criterion addressed?	3
Is the recommendation well-justified?	2

Is the report clear and unbiased? 2

Task 7 — Display Output on UI

What happens: The UI shows the final comparison report, scores, and recommendation in a clean, readable format.

Input

• Comparison table from Task 5

• Score + feedback from Task 6

Output (what the user sees)

• Side-by-side comparison table on the webpage

• Ranked leaderboard with scores

• Pros & Cons cards per tool

• Judge score shown as a number with a label

• Feedback shown as a small text box below the table

• Export button to download as PDF or CSV

Full Flow Summary

User fills comparison form

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Agent shortlists AI tools

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Agent searches for feature & pricing data

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Agent scores & ranks tools

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Agent generates comparison report

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LLM-as-Judge scores the report

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Output displayed on UI

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Deployed live on Railway / Vercel